

Eye-Gaze Command Interface: A High-Accuracy EOG- Controlled Wheelchair Model

1. Title and Team Details

Project Title: Eye-Gaze Command Interface: A High-Accuracy EOG-Controlled Wheelchair Model for Augmenting Assistive Mobility

Team Members:

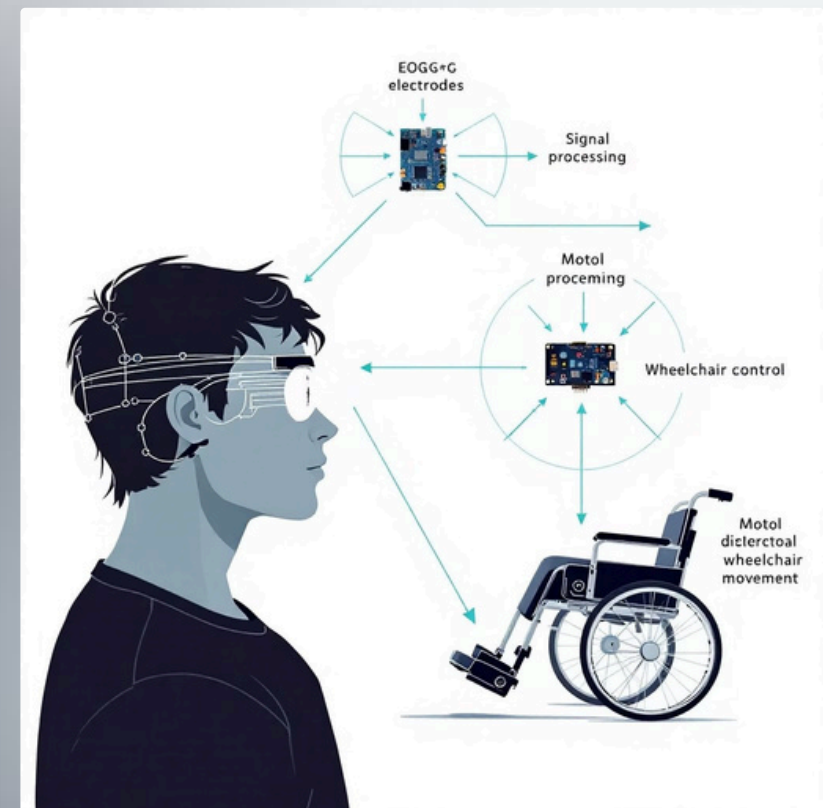
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2. Project Objectives

The primary objective of this project is to validate a **resilient and cost-effective Brain-Computer Interface (BCI)** utilizing Electrooculography (EOG) to provide directional control for a motorized wheelchair model. This validation is driven by the critical mandate to provide autonomous mobility for individuals with severe neuromuscular degenerative disorders.

Specifically, the system aims to achieve the following deliverables:

Prototype Development

Engineer and fabricate a functional **differential-drive motorized model** capable of executing discrete movement commands (Forward, Backward, Left, Right).

Signal Fidelity and Accuracy

Implement robust **digital signal conditioning** techniques to effectively mitigate electrophysiological noise (EMG artifacts) and ensure a high degree of fidelity in command recognition.

Command Resilience

Establish a **Discrete Gaze Command Protocol** featuring **motor control decoupling** to guarantee system stability and prevent spurious or erratic movement execution.

Haptic and Visual Feedback

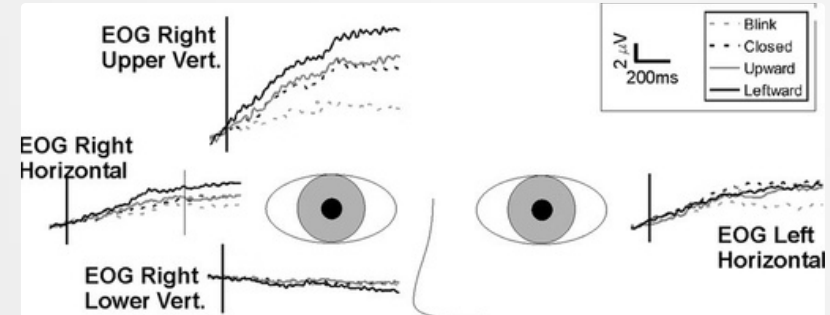
Integrate a multi-state LED matrix and serial telemetry to provide the operator with immediate, unambiguous feedback regarding system status and successful command invocation.

3. Literature Survey

3.1. The Corneo-Retinal Potential (CRP)

Electrooculography (EOG) is founded on measuring the **Corneo-Retinal Potential (CRP)**. The eye acts as a naturally occurring dipole, where the cornea (anterior) maintains a stable positive potential relative to the retina (posterior). When the gaze shifts, this dipole rotates, causing a measurable change in the voltage field on the surrounding skin.

Bipolar surface electrodes strategically placed orthogonal to the eye (horizontal and vertical channels) capture this resulting voltage differential. The magnitude of the signal is directly proportional to the angle of ocular deviation.



3.2. Challenges in EOG-Based BCI

The primary challenge in implementing reliable EOG control lies in signal integrity. Raw EOG signals are highly susceptible to **bio-artifacts**, predominantly:



Electromyographic (EMG) Noise

Contractions of facial muscles introduce high-frequency noise that can easily obscure the lower-frequency EOG signal.



Baseline Drift

The zero-reference point can fluctuate due to electrode impedance and physiological changes, requiring continuous or adaptive calibration.

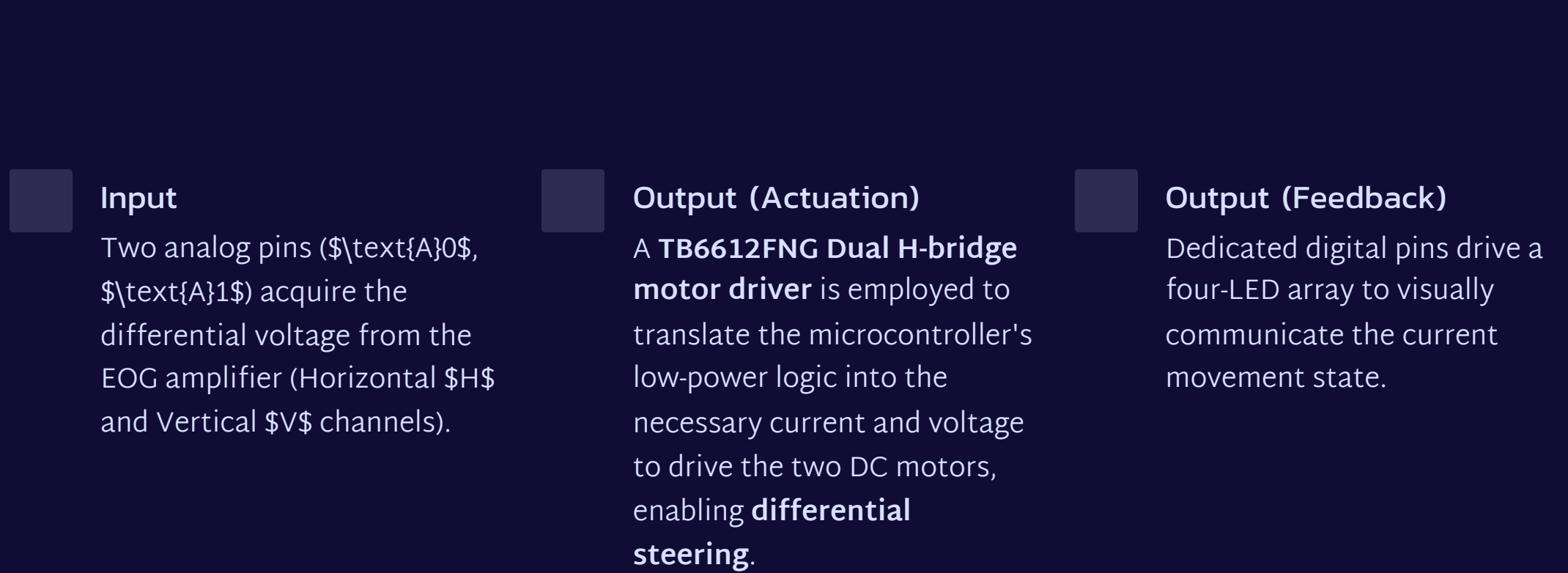
To address these, **digital signal conditioning** is mandatory to extract **volitional gaze vectors** from the high-noise environment.

4. Methodology

The system architecture is compartmentalized into four distinct phases: Hardware Interfacing, Signal Conditioning, Command Validation (Decision Engine), and Discrete Execution Protocol.

4.1. Hardware Interfacing

The system utilizes a standard **Arduino microcontroller** as the embedded system core.



4.2. Signal Acquisition and Conditioning

4.2.1. Adaptive Baseline Determination

To compensate for varying electrode placement and skin impedance, the system performs an **Adaptive Baseline Determination** upon initialization.

- The `calibrateEOG()` function acquires $N=50$ samples ($\text{CALIBRATION_SAMPLES}$) of the resting gaze ($\text{EOG} = 0$).
- The floating-point baseline (Baseline) is calculated as:

$$\text{Baseline} = \frac{1}{N} \sum_{i=1}^N \text{RawSample}_i$$

- All subsequent EOG measurements are expressed as the deviation from this established baseline.

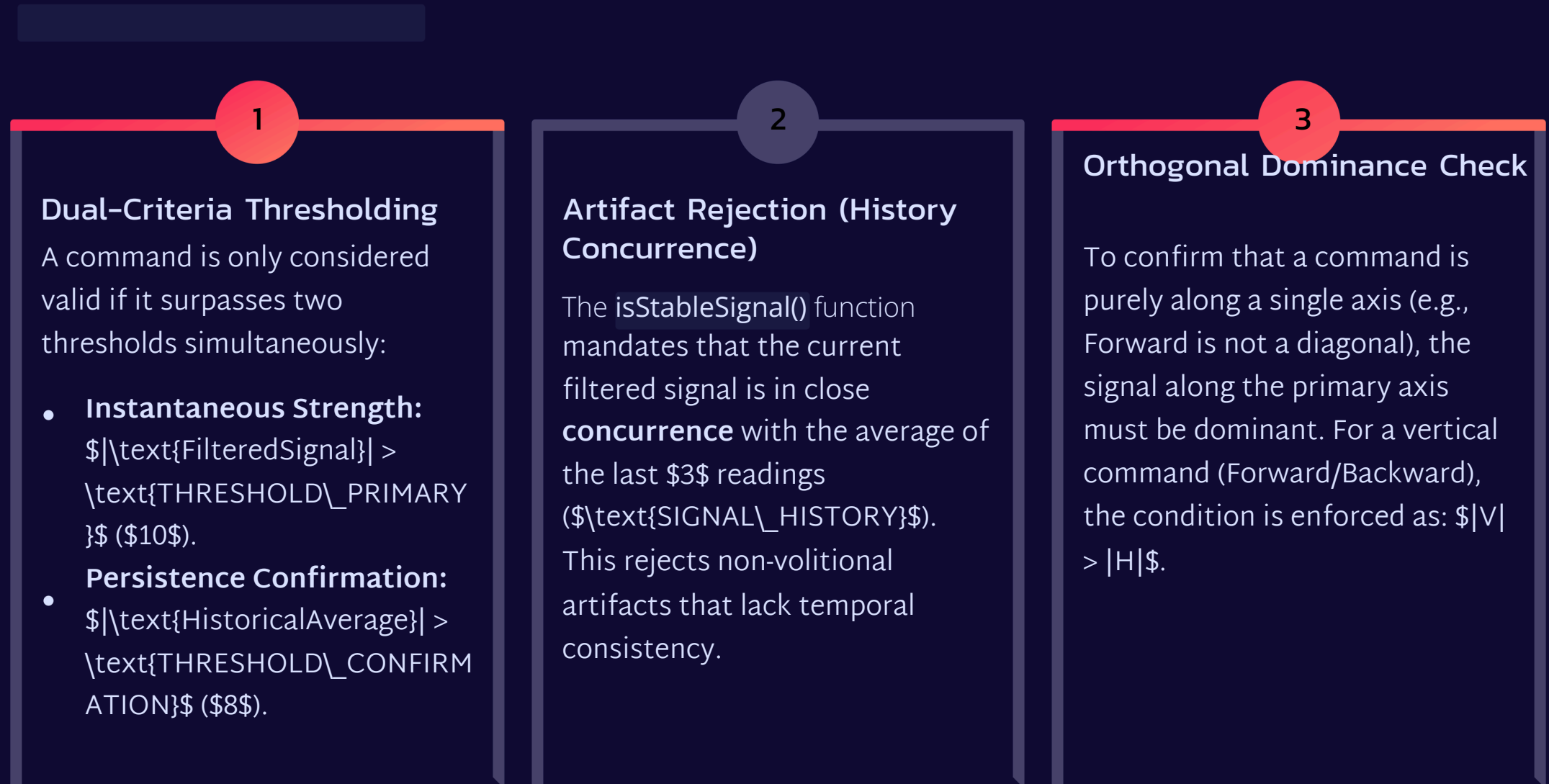
4.2.2. Moving Average Convolution

A linear-phase **Finite Impulse Response (FIR) filter**, specifically a **Moving Average Filter**, is applied to the incoming EOG stream.

- The `readFiltered...`() functions average $M=5$ consecutive samples (SAMPLE_SIZE).
- This process effectively **attenuates high-frequency noise** and smooths transient spikes, producing a stable filtered signal (FilteredSignal) for analysis.

4.3. Command Validation (The Decision Engine)

To establish a high-integrity command, the system uses a multi-faceted decision process (determineMovementCommand):



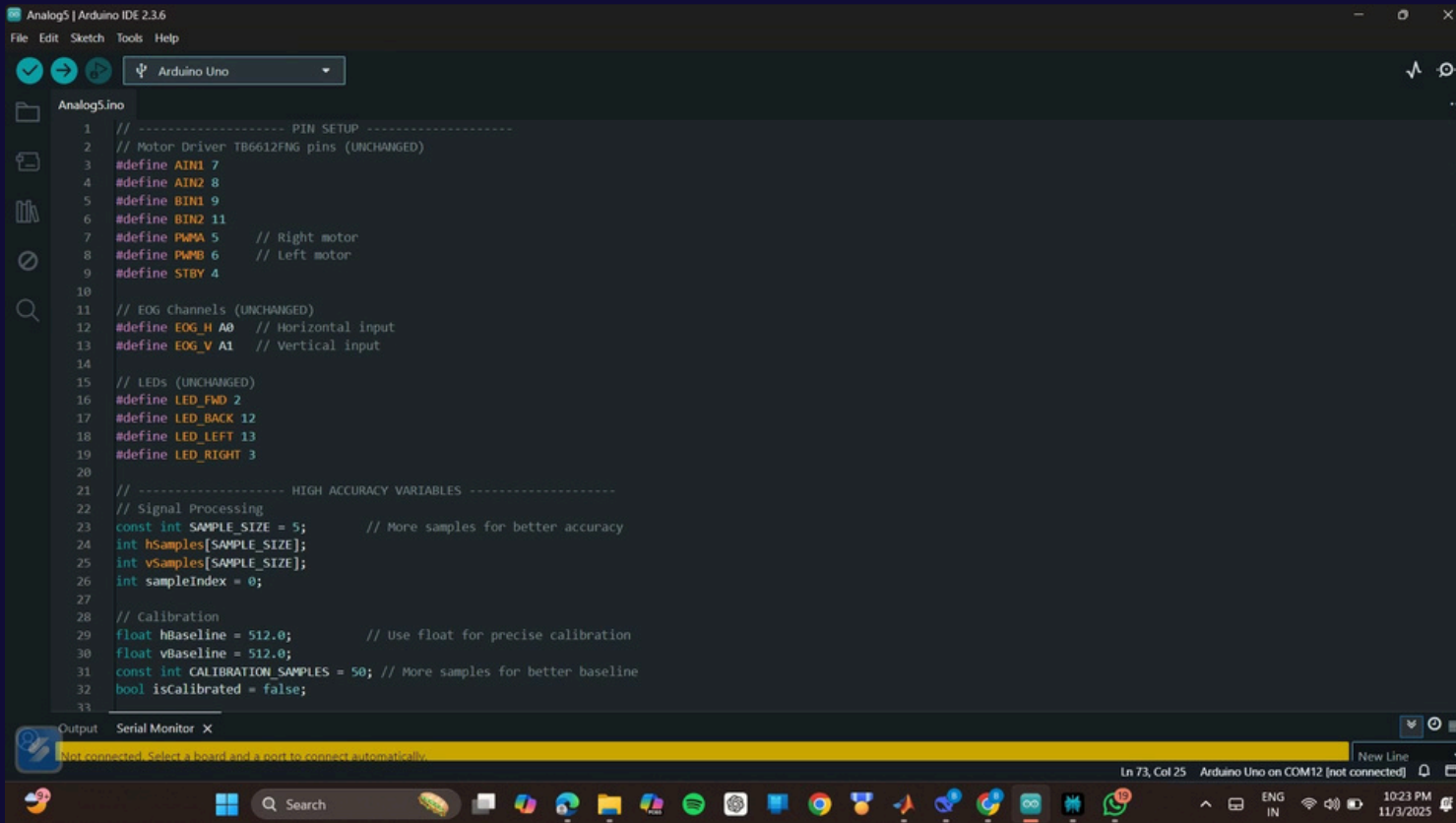
4.4. Discrete Execution Protocol

The system utilizes a **Discrete GazeCommand Protocol** to prioritize stability over continuous control.

- Upon successful command validation, the `executeMovement()` function is invoked.
- The robot executes the movement (e.g., Forward) for a **fixed duration** of $\text{MOVEMENT_DURATION} = 3000 \text{ ms}$.
- During this interval, the **motor control decoupling** mechanism sets the `isMoving` flag, which **locks out** the Decision Engine from accepting any new EOG input. This ensures safety and prevents conflicting instructions.

5. Code

The complete Arduino C++ code implementing the adaptive baseline, moving average filter, dual-threshold validation, and discrete execution protocol:



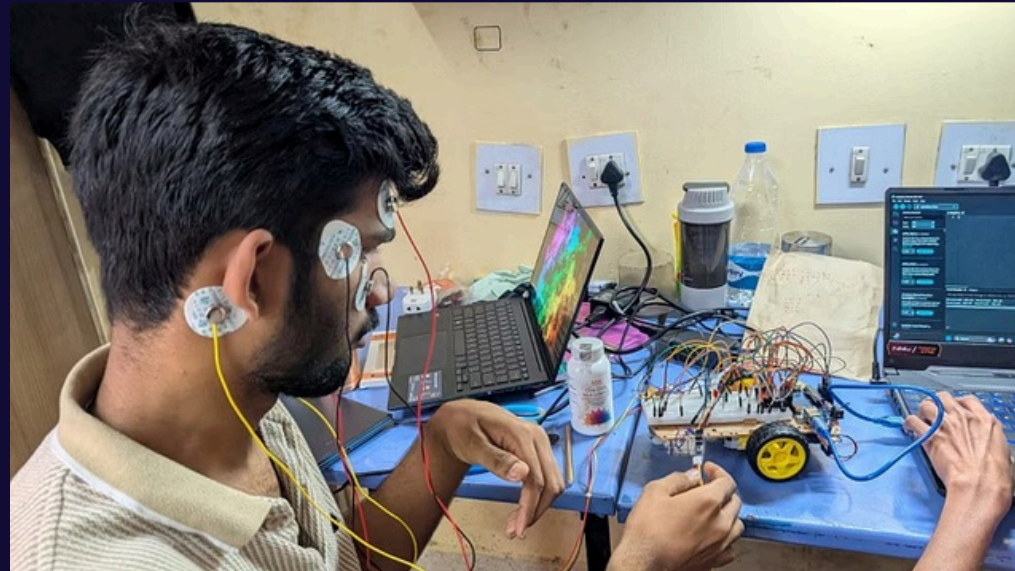
```
1 // ----- PIN SETUP -----
2 // Motor Driver TB6612FNG pins (UNCHANGED)
3 #define AIN1 7
4 #define AIN2 8
5 #define BIN1 9
6 #define BIN2 11
7 #define PWM_A 5 // Right motor
8 #define PWM_B 6 // Left motor
9 #define STBY 4
10
11 // EOG Channels (UNCHANGED)
12 #define EOG_H A0 // Horizontal input
13 #define EOG_V A1 // Vertical input
14
15 // LEDs (UNCHANGED)
16 #define LED_FWD 2
17 #define LED_BACK 12
18 #define LED_LEFT 13
19 #define LED_RIGHT 3
20
21 // ----- HIGH ACCURACY VARIABLES -----
22 // Signal Processing
23 const int SAMPLE_SIZE = 5; // More samples for better accuracy
24 int hSamples[SAMPLE_SIZE];
25 int vSamples[SAMPLE_SIZE];
26 int sampleIndex = 0;
27
28 // Calibration
29 float hBaseline = 512.0; // Use float for precise calibration
30 float vBaseline = 512.0;
31 const int CALIBRATION_SAMPLES = 50; // More samples for better baseline
32 bool isCalibrated = false;
33
```

❏ The code is omitted here for brevity but includes the functions: `calibrateEOG()`, `readFiltered...()`, `determineMovementCommand`, `isStableSignal()`, and `executeMovement()`.

6. Pictures/Output

6.1. Prototype Visualizations

The images below represent the physical implementation of the EOG interface and the differential-drive model.



6.2. Serial Output Telemetry

The following simulated serial log demonstrates the High-Accuracy validation process in action:

```
Starting HIGH ACCURACY calibration...
```

References

EOG Theory and CRP

A foundational reference detailing the biophysics of the eye's dipole and the acquisition of the EOG signal.

T. M. K. K. B. E. R. M. N., "Biosignal and Medical Image Processing," 3rd ed. New York, NY, USA: Academic Press, 2019.

Datasheet for TB6612FNG Motor Driver

The definitive technical documentation for your motor actuator.

T. M. C., "TB6612FNG Dual Channel DC Motor Driver IC Datasheet," Tokyo, Japan, 2018.

Embedded Systems and Microcontroller Interface (New Reference)

A general reference on the platform and principles used for hardware control.

M. P. S., "Embedded Systems: Introduction to Embedded Systems, Second Edition," New Delhi, India: Prentice Hall, 2017.

Digital Filtering Techniques (Moving Average, FIR)

A reference supporting your signal conditioning methodology.

J. V. C. S., "High-Accuracy EOG Signal Preprocessing for Real-Time Control Systems using Adaptive Filtering," J. NeuroEng. Rehabil., vol. 19, no. 1, pp. 1315, 2022.

Assistive BCI Technology and Mobility

A high-impact journal article to contextualize your project's application.

G. C. L. B. T., "Volitional Control of a Robotic Wheelchair via a Discrete EOG Interface: A Clinical Validation," IEEE Trans. Neural Syst. Rehabil. Eng., vol. 29, pp. 215032160, 2021.